

Orthogonal Norm Growth in Two-Tower Retrieval: A Popularity Mechanism for Unified Discovery

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Abstract

Unified search and recommendation systems may update the same item vectors from both interaction streams. We study how repeated updates can increase an item vector’s norm and affect retrieval under an unnormalized dot product. A cosine-only loss makes the output gradient perpendicular to the current item vector—tangent to its constant-norm sphere. Training changes encoder parameters, however, so the realized item-vector movement need not remain tangent. We derive sufficient conditions under which it does; every nonzero update then increases squared norm by its squared length. When the tasks share an item encoder, sequential updates from both can accumulate in the same norm, which raw-inner-product serving can expose for either task. This conditional result is not a theorem about joint SGD.

Controlled ablations test these conditions. On MovieLens 32M and Amazon Books, accumulated tangent movement accompanies frequency–norm association and additional head-item exposure under raw-dot rather than normalized-item ranking. In a nonlinear tower combining a per-item ID vector with a shared content encoding, active item movement remains near-tangent. We separate its norm change into the ID-update contribution and other output changes, then use a matched training intervention that removes immediate norm growth caused by the ID-row update. The intervention reduces norm-related exposure but also recommendation utility, so it is diagnostic rather than a proposed mitigation. Our logs are recommendation-only: the formulation supplies a shared-representation audit, while prevalence in joint traffic remains open.

Keywords

unified search and recommendation, neural retrieval, two-tower models, embedding norms, popularity bias, representation geometry

1 Introduction

Search and recommendation are complementary routes to a shared catalog: search starts from an explicit query, whereas recommendation infers an information need from a user and context. Unified systems connect their behavior sequences, supervision, model parameters, or item representations [10, 15, 19]. At candidate-generation scale, both tasks commonly use bi-encoder or two-tower retrieval: a request-side encoder produces a query or user vector, an item-side encoder produces a catalog vector, and a simple similarity enables precomputation and approximate nearest-neighbor search [5, 6]. Sharing the item tower is efficient, but it also lets training on one

surface change the representation served on the other. Consider an item that receives repeated search-side updates. Under the conditions derived below, those updates add to the squared norm of the same catalog vector later used by recommendation. If recommendation retrieves with raw inner products, this search-derived magnitude enters its scores even if its request tower is unchanged; normalizing item vectors before serving removes this particular channel. This cross-surface dependency motivates measuring norm growth separately by task.

We study the norm component of this cross-surface dependency. If item frequency becomes correlated with item-vector norm, raw-inner-product ranking contains a frequency-associated component:

$$\langle u, q_i \rangle = \|u\| \|q_i\| \cos(u, q_i). \quad (1)$$

That component may improve utility, amplify exposure, or do both; popularity is not intrinsically harmful [9, 16]. Our question is mechanistic: *when can repeated updates under an objective whose item-output dependence is only through cosine similarities accumulate item norm, how can two tasks contribute to the same norm, and when does that norm affect retrieval exposure?*

Throughout, the *output gradient* is the derivative with respect to the current item vector q_i , whereas the *realized displacement* Δq_i is its before–after change caused by an optimizer step. We call a gradient or displacement *tangent* at q_i when its inner product with q_i is zero, meaning that the two vectors are perpendicular. A common argument is that a cosine objective has a tangent output gradient, so the item vector itself must move tangentially and increase its norm [2, 3, 18]. A deep encoder instead updates parameters; its Jacobian can rotate the output gradient and mix it with gradients from other examples. We therefore distinguish

$$q_i^\top \nabla_{q_i} \mathcal{L} = 0 \quad \text{from} \quad q_i^\top \Delta q_i = 0,$$

and identify sufficient conditions under which the first statement does imply the second.

The theorem below gives narrow, testable sufficient conditions rather than a description of every modern tower. Outside those conditions, tangent output motion must be measured rather than assumed. We therefore also study a nonlinear item tower that sums a learned per-item ID vector and a shared content encoding. The optimizer rescales the active ID row’s gradient by one scalar, so the output change due to that row remains exactly tangent; the shared content update supplies the residual that can break tangency. Tangency of the total item-vector movement is then a measurable property rather than a guarantee.

Our contributions are:

- **Update-level theory.** We derive the parameter-to-output update and prove sufficient conditions for exact tangent item motion, showing when a tangent output gradient does—and

does not—imply tangent movement of a parameterized item vector.

- **Shared-stream accumulation.** Under uniform interaction sampling and conditional on trajectory-dependent increments, sequential task-labeled search and recommendation updates contribute additively to one shared item norm. We state explicitly what this accounting does and does not prove about joint SGD or mixed-task minibatches.
- **Mechanism-oriented evidence.** On MovieLens 32M and Amazon Books, condition tests, total-output diagnostics, exact path accounting, and a matched intervention connect tangent motion to frequency-associated norm growth and norm-induced head exposure, while reporting the accompanying utility cost.
- **Unified-discovery audit.** We translate the mechanism into task-separated diagnostics for shared item towers and serving indices. Our experiments use recommendation logs, so search-side prevalence remains a scoped validation target rather than a claimed result.

Artifacts. Code, configurations, and saved outputs are in the repository (click here). Full derivations, expanded protocols, confidence intervals, and diagnostic tables are in the supplementary document (click here). The paper contains the complete claims and principal evidence.

2 Positioning

Unified search and recommendation. Joint search–recommendation research has integrated heterogeneous behavior sequences [19], introduced paired real-world logs [15], and shared scenario representations and supervision in industrial ranking systems [10]. These works ask how information should transfer between tasks. We study a complementary question: what optimizer-induced state can be transferred unintentionally when an item representation or retrieval index is shared. The result is not a new unified architecture; it is an audit of a coupling channel that such architectures may contain.

Two-tower retrieval and item representations. Two-tower models support large-scale web search [6] and recommender candidate generation [1, 5]. A request tower may be deep and sequential [14], while an item side can remain a shallow ID matrix because such representations are inexpensive to precompute and can be effective [20]. Our exact conditions apply per tower: complexity on the request side neither establishes nor invalidates the item-side guarantee.

Cosine geometry and popularity. Normalized objectives are known to produce output gradients tangent to the hypersphere [18]. Recent analyses connect this fact to norm growth [2, 3]. We refine the connection by separating the output gradient from the realized output displacement through the encoder Jacobian. In recommendation, popularity bias has multiple data-, model-, and evaluation-level causes [9]; representation collapse provides another model-side explanation [21]. We isolate one frequency-associated magnitude path. Kim et al. report positive correlations between item popularity and embedding norm, decompose inner-product scores into angular

and magnitude factors, and introduce partial or full item normalization at inference [8]. We therefore do not claim the association, the algebraic score decomposition, or unit-item rescoring as novel. Our distinct contribution is upstream: we analyze the parameter-to-output map, derive sufficient conditions for tangent realized item motion, and account for that motion across task-labeled updates of a shared item representation. We do not claim this path is necessary, dominant, or harmful in every system.

Optimization and serving boundary. Contrastive objectives such as InfoNCE [17] determine the output gradient, but the optimizer determines how that gradient changes a parameterized representation. Sparse recommendation systems consequently use specialized row-wise adaptive updates [7], while dense branches commonly use coordinate-wise adaptive optimizers. Our theorem covers direction-preserving row scalings; the hybrid study makes the residual from a dense shared branch explicit. Separately, training with normalized scores does not force serving to discard magnitude. Large-scale two-tower infrastructure may consume raw inner products for candidate retrieval or pass embeddings to a later scorer [5]. We therefore treat raw-versus-unit ranking as a serving decomposition, not as an assumption that every deployed system deliberately mismatches its training score.

3 A Shared-Tower Norm Mechanism

3.1 Two streams, one item representation

We use *stream* for the source of a training event and *surface* for a search or recommendation serving endpoint. Let $a \in \{S, R\}$ denote the search and recommendation streams. A task-specific request encoder produces u^a , while an item encoder $q_i = q_\theta(x_i)$ is shared. Training events from stream a compare u^a with positive and negative item outputs through an objective whose dependence on those outputs is only through cosine similarities. Under uniform-over-events sampling within a stream, item i has event probability p_i^a . Events from the two streams may be interleaved in any schedule; the argument below concerns the item path at each realized update.

At serving, either surface may use unit-item scoring, which excludes item magnitude, or a raw inner product, which includes the factor $\|q_i\|$ in Equation (1). Thus shared training alone does not make norm affect rankings: a serving path must also admit magnitude.

3.2 The missing parameter-to-output map

Let $\mathcal{I}_B$ be the distinct item IDs that appear as positives or negatives in a minibatch. For each $\ell \in \mathcal{I}_B$, let q_ℓ be its current item-tower output, $J_\ell = \partial q_\ell / \partial \theta$, and $\bar{g}_\ell$ the sum of the output-gradient contributions from all occurrences of that item ID in the minibatch. An SGD step gives

$$\begin{aligned} \Delta \theta &= -\eta \sum_{\ell \in \mathcal{I}_B} J_\ell^\top \bar{g}_\ell, \\ \Delta q_i &\approx -\eta \sum_{\ell \in \mathcal{I}_B} J_i J_\ell^\top \bar{g}_\ell. \end{aligned} \tag{2}$$

We call the output *parameter-linear* when it is linear in the parameters updated by the step; then the second relation is exact for a finite step. In a general encoder it is only a first-order approximation. The equation exposes two obstacles: the self-map $J_i J_i^\top$ may rotate

$\bar{g}_i$, and the cross-item maps $J_i J_\ell^\top$ may inject gradients belonging to other item IDs.

We call an item encoder *lookup-equivalent* when different item IDs have independent parameter blocks and the Jacobian products satisfy

$$J_i J_\ell^\top = \begin{cases} c_i I, & i = \ell, \\ 0, & i \neq \ell, \end{cases} \quad c_i > 0, \quad (3)$$

where I is the output-space identity matrix. In plain terms, updating one item's parameter block cannot move another item's output, and the item's own parameter-to-output map can rescale but cannot rotate its output gradient. A standard embedding table has $c_i = 1$. A bias-free linear map applied to one-hot item IDs has the same Jacobian property and is therefore lookup-equivalent under this definition. By contrast, a linear map on nonorthogonal features has $J_i J_\ell^\top = (x_i^\top x_\ell) I$, so cross-item terms survive. Two consecutive linear layers are linear in their input but not jointly linear in their parameters, so Equation (2) is no longer exact for a finite step.

The vector $\bar{g}_i$ already aggregates repeated occurrences of item ID i . We call an item-block update *direction-preserving* when it multiplies the entire current parameter-block gradient $J_i^\top \bar{g}_i$ by one positive scalar shared across coordinates. Plain SGD and Exact Row-Wise Adagrad are examples. Momentum, weight decay, coordinate-wise preconditioning, shared nonorthogonal features, or nonlinear shared parameters remove this guarantee; they do not imply that empirical motion must be non-tangent.

3.3 Exact tangent motion

A loss that depends on nonzero q only through cosine similarities is invariant to positive rescaling: $\mathcal{L}(\alpha q) = \mathcal{L}(q)$ for $\alpha > 0$. Differentiating at $\alpha = 1$ gives

$$q^\top \nabla_q \mathcal{L} = 0. \quad (4)$$

THEOREM 3.1 (SUFFICIENT CONDITION FOR TANGENT ITEM MOTION). *Suppose an item output is produced by a parameter-linear, lookup-equivalent encoder; the loss depends on that output only through cosine similarities; and its item-specific parameter block is updated by a positive scalar multiple of the current block gradient, without momentum or an additional regularization term. Then the realized item update is tangent: $q_i^\top \Delta q_i = 0$.*

PROOF. Every occurrence of item ID i is evaluated at the same pre-step output q_i . Equation (4) therefore gives $q_i^\top \bar{g}_i = 0$. Lookup-equivalence removes contributions from other item IDs. For some block-update scalar $s_i > 0$, the direction-preserving rule gives $\Delta \theta_i = -s_i J_i^\top \bar{g}_i$. Parameter-linearity and Equation (3) then give the exact output change

$$\Delta q_i = J_i \Delta \theta_i = -s_i J_i J_i^\top \bar{g}_i = -s_i c_i \bar{g}_i.$$

It follows directly that $q_i^\top \Delta q_i = 0$. $\square$

Expanding the squared norm makes the resulting equality explicit:

$$\begin{aligned} \|q_i^{(t+1)}\|^2 - \|q_i^{(t)}\|^2 &= 2(q_i^{(t)})^\top \Delta q_i^{(t)} + \|\Delta q_i^{(t)}\|^2 \\ &= \|\Delta q_i^{(t)}\|^2 \geq 0. \end{aligned} \quad (5)$$

No large learning rate is required for this identity. Detectability depends on cumulative tangent energy $\sum_t \|\Delta q_i^{(t)}\|^2$ relative to initial norm variation and non-tangent trajectory components.

3.4 Conditional accumulation across tasks

For item i , let $N_i^a(T)$ be its number of update events originating from stream a by time T , and let $\delta_{i,k}^a = \|\Delta q_{i,k}^a\|^2$. If every interleaved item update satisfies Theorem 3.1, summing the exact per-step identity in Equation (5) and partitioning increments by their source stream gives the pathwise equality

$$\|q_{i,T}\|^2 = \|q_{i,0}\|^2 + \sum_{k=1}^{N_i^S(T)} \delta_{i,k}^S + \sum_{k=1}^{N_i^R(T)} \delta_{i,k}^R. \quad (6)$$

Condition on both nonnegative increment sequences. Coupling event draws with common uniforms shows that the expected squared norm is nondecreasing in each p_i^S and p_i^R separately: a higher-probability stream contains every event of its lower-probability counterpart and possibly more.

This accounting formalizes a shared-state channel: opportunities from either task enter one magnitude that either raw-inner-product surface can consume. It does *not* prove that two items are ordered by popularity, that changing one stream leaves either increment sequence fixed, or that joint training improves or harms either task. In full SGD, frequencies affect batches, negatives, gradients, and the shared trajectory. We therefore test below whether opportunity count or per-update energy dominates in a less constrained tower, and reserve search-side prevalence for joint-log validation.

Equation (6) assumes task-labeled updates are applied sequentially. If one minibatch mixes both tasks, their aggregated tangent components still yield a tangent total update, but its squared length contains a cross-task inner product. It should then be recorded as one combined increment unless the optimizer exposes separate task components; assigning that cross term to either stream is not identified by norm accounting alone.

4 Mechanism-Oriented Experiments

4.1 Questions, data, and measurements

The experiments follow the argument rather than benchmark architectures: (Q1) do realized updates obey the predicted geometry when the sufficient conditions hold, and how do they change when individual conditions are violated; (Q2) does sufficient cumulative tangent motion produce frequency–norm association and magnitude-dependent retrieval; and (Q3) can a traceable tangent path remain active in a nonlinear tower?

Experiments 0–7 use a hand-built graph of 25 positive interactions among 7 users and 19 items. Experiments 8–10 use MovieLens 32M [4] (32,000,204 interactions, 200,948 users, 84,432 interacted movies) and Amazon Books [11]. A full-data joint 3-core of Amazon retains 35,179,026 interactions, 3,788,815 users, and 1,214,141 items. Ratings are implicit positives.

For geometry, we record outputs immediately before and after an optimizer step and measure $|\cos \angle(q_i, \Delta q_i)|$. For association, we use Pearson correlation between training frequency and final item norm. For serving, after masking a user's training history, we report the fraction of the top 100 occupied by the most frequent

Table 1: Controlled tests of the sufficient conditions. “Tangent” refers to realized output motion, not merely the output gradient.

Exp.	Encoder diagnosed	Objective/update	Tangent?
0	embedding	cosine; momentum/Adam	no
1	embedding	cosine; SGD	yes
2	embedding	dot product; SGD	no
3	deep user / item lookup	cosine; Adam/SGD	no/yes
4	embedding + linear	cosine; SGD	no
5	fixed features + linear	cosine; SGD	no
6	one-hot + linear	cosine; SGD	yes
7	embedding	$(1 - \cos)^2$; SGD	yes

1% of training items. At its fully normalized endpoint, the test-time method of Kim et al. induces the same per-user ranking as our unit-item score [8]. We compare this endpoint with raw inner product inside one trained model to diagnose the contribution admitted by item magnitude, not to introduce a mitigation method. This comparison is not an endorsement of a train/serve mismatch and does not label either ranking distribution as ideal.

4.2 Testing the sufficient conditions

Experiments 0–7 use cosine InfoNCE with in-batch negatives [17], plain SGD unless stated otherwise, and a lookup item tower. Table 1 varies one optimizer, objective, or encoder property at a time. Configurations satisfying all sufficient conditions are tangent at every observed update. Momentum 0.9, Adam, a dot-product objective, a shared two-layer map, and a linear map on nonorthogonal features produce non-tangent trajectories in these tests. These violations demonstrate loss of the guarantee in concrete cases; they do not establish that each condition is necessary for every trajectory. For the shared two-layer and nonorthogonal-feature cases, only about 20% of observed angles fall within $90^\circ \pm 10^\circ$, and the trajectories span the full half-circle rather than concentrating around tangency.

Experiments 8 and 9 use two 100-dimensional embedding towers, the only difference is bigger and smaller learning rate, respectively. Full protocol is in repository (click here). Their contrast probes cumulative displacement, not a portable nominal learning-rate threshold. Experiment 10 uses a BERT-like sequential user tower [14] with Adam and weight decay, and a disjoint 100-dimensional item lookup with SGD and a cyclic schedule [13]; only the item side lies in the exact regime.

Tangency holds in both Experiments 8 and 9, but only the larger accumulated path overcomes initialization-scale norm variation. Frequency–norm correlations in Experiments 8 and 10 reject a zero-correlation null at $p < .05$ on both datasets; Experiment 9 remains negligible. Thus exact tangency is sufficient for per-step norm increase but not for a detectable cross-item correlation. When correlation emerges, magnitude can substantially raise head exposure under raw-dot ranking. The unit-item column may itself contain relevance, log-exposure, or other popularity effects; only the within-model gap belongs to the magnitude channel diagnosed here. These runs are mechanism stress tests, not quality-optimized recommenders.

4.3 A nonlinear tower beyond the guarantee

We next study

$$q_i = e_i + \alpha f_\theta(x_i), \quad (7)$$

where the ID row e_i uses no-weight-decay Exact RowWise Adagrad [7, 12], and a shared two-layer GELU content encoder uses AdamW with weight decay. The complete tower is nonlinear and shares parameters across items. For an active item, let $d_{i,t}$ denote the component of its output change caused by updating its ID row. Exact RowWise Adagrad rescales the whole row gradient by one scalar, so this component is exactly tangent to the current total item vector. Writing the simultaneous change caused by the shared content encoder as $r_{i,t}$ gives $\Delta q_{i,t} = d_{i,t} + r_{i,t}$ and

$$|\cos \angle(q_{i,t}, \Delta q_{i,t})| \leq \frac{\|r_{i,t}\|}{\|\Delta q_{i,t}\|}. \quad (8)$$

Tangency of the total item-vector movement is now empirical, but any deviation has a localized source.

Protocol. We use global temporal 80/20 splits, fit user/item 3-cores and all popularity statistics on training only, mask train-seen items, and evaluate warm users with at least three future positives. MovieLens content contains genres and release-decade buckets; Amazon uses hashed category, brand, and date metadata with explicit missing values. A 256-dimensional hidden layer maps content into the 128-dimensional item space; users are 128-dimensional ID vectors. Full protocol is in repository (click here).

For each item, define total squared-norm change $G_i = \|q_{i,T}\|^2 - \|q_{i,0}\|^2$, accumulated tangent energy $D_i = \sum_t \|d_{i,t}\|^2$, and signed residual $H_i = G_i - D_i$. The identity $G_i = D_i + H_i$ is exact pathwise. With $z_i = \log(1 + f_i)$, define $C_X = \text{Cov}_i(z_i, X_i)$. The signed ratio C_D/C_G localizes a frequency-associated term but is not a causal percentage.

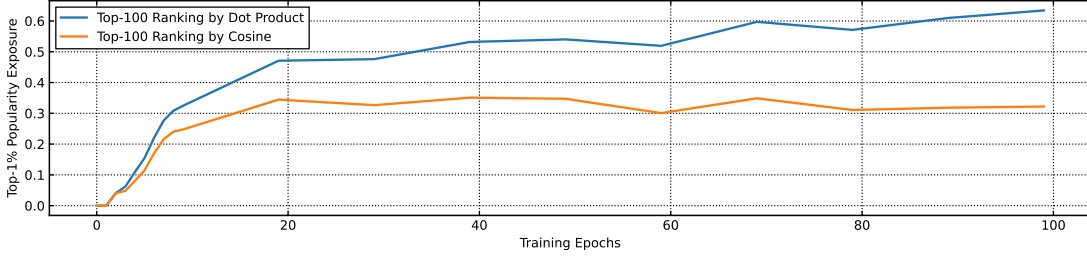
The angles of total active item-vector movements are much closer to tangency than an isotropic 128-dimensional reference (median/q95 = .0599/.1729). The content function is not approximately frozen: removing it worsens held-out loss in every seed, although fine-tuning it over its frozen initialization helps only on Amazon. The exact covariance closure holds in every run. On MovieLens, the residual opposes the positive tangent covariance; on Amazon it reinforces it, while the tangent term remains larger.

Most directly for Equation (6), log-frequency correlates with ID update count but not positively with tangent energy per update (Table 3). Accumulation therefore follows additional opportunities, even though frequent Amazon items receive smaller tangent energy per update.

The tangent-only counterfactual and learned norms raise head exposure on both datasets, but standard norm lift is dataset-dependent: 1.62 points on MovieLens and 20.12 on Amazon. Retracting each ID-row update’s immediate norm change lowers frequency–norm correlation in all six runs and shifts norm lift downward; paired user-bootstrap 95% intervals for the lift reduction exclude zero in every run. Raw-dot Recall and NDCG also decrease in every run, by roughly 17–22% relative, with paired intervals excluding zero. Unit-item utility changes are much smaller and mostly inconclusive. Retraction therefore tests sensitivity to immediate norm growth caused by ID-row updates; it is neither a norm-only post-hoc edit nor a cost-free mitigation.

Table 2: Exact-regime outcomes. Paired entries are MovieLens 32M/Amazon Books. Head share is the top-1%-frequency fraction of the top 100. Dot-unit differences diagnose item magnitude within a model.

Exp.	Request-side tower	Tangent item path	Frequency-norm r	Top-1% dot	Top-1% unit item
8	embedding; larger cumulative motion	yes	.66/.63	.72/.60	.29/.06
9	embedding; small cumulative motion	yes	.0055/.0002	.007/.010	.007/.010
10	BERT-like; Adam + weight decay	yes	.57/.61	.63/.98	.32/.11

**Figure 1: Experiment 10 on MovieLens 32M. Item norms and raw-dot head share increase as tangent item updates accumulate, while unit-item ranking excludes magnitude. The trajectories visualize association, not causal mediation of all head exposure by norm.****Table 3: Hybrid mechanism diagnostics (means across three seeds; selected sample SDs shown). Arrowed r is initial→final→retracted frequency-norm correlation.**

Metric	ML-32M	Amazon
Active cos median/q95	.0034/.0426	.0003/.0095
$\ r\ /\ \Delta q\ $ median/q95	.0166/.0972	.0019/.0384
Frequency-norm r	-.002 → .465 → .199	-.000 → .404 → .148
C_D/C_G	1.104 ± .002	.686 ± .001
$r(z, N)$	.569	.439
$r(z, D/N)$	-.032	-.629
Content function change M_f	2.420 ± .087	1.467 ± .072

5 Unified-Discovery Audit and Limits

Equation (6) makes the unified-system implication specific. If search and recommendation update one direction-preserving item row, their tangent energies accumulate in one norm. If either surface then uses a shared raw-inner-product index or passes magnitude to a downstream scorer, state learned from one stream can affect candidates on the other. Neither joint training nor a shared catalog is sufficient by itself: the relevant item parameters must be shared, the update path must retain a tangent component, and serving must admit magnitude.

This motivates four task-separated checks:

- (1) **Measure realized motion.** Log $q_i^T \Delta q_i / (\|q_i\| \|\Delta q_i\|)$ after optimizer steps. Logging $q_i^T \nabla_{q_i} \mathcal{L}$ alone tests only output-gradient geometry.
- (2) **Attribute opportunities and energy.** Maintain N_i^S, N_i^R and accumulated D_i^S, D_i^R , then report count and energy per event by task and frequency bucket. This distinguishes more opportunities from larger updates.
- (3) **Decompose serving.** Within each trained model and surface, compare raw inner product with unit-item scoring after

identical candidate masking. Their difference diagnoses magnitude; it does not define unbiased relevance.

- (4) **Test cross-task sensitivity.** Compare search-only, recommendation-only, and joint updates of the shared item tower, followed by both serving surfaces. A matched retraction can test trajectory sensitivity, but its utility must be reported and it should not be described as pure mediation.

What transfers mathematically. The tangent-update and additive-accounting statements are agnostic to whether a realized event came from an explicit query or an implicit recommendation context. Consequently, a lookup-equivalent shared row satisfying the theorem has the same per-update norm law in either stream. This is the supported cross-task result.

What remains empirical. Our datasets contain recommendation interactions, not paired search and recommendation logs such as KuaiSAR [15]. Query traffic can change item frequencies, negative construction, relevance supervision, request directions, and score calibration. We therefore do not know whether the tangent path is prevalent or material in a deployed joint model, whether search or recommendation contributes more energy, or whether the induced magnitude improves either task. The proposed task-separated experiment is future validation, not evidence reported here.

Other limitations. The exact theorem is sufficient and covers a narrow direction-preserving row path; it is not a characterization of AdamW-trained shared MLPs. The hybrid experiment establishes existence beyond the exact regime for one construction, not prevalence across architectures. Frequency is observational in the data, and $G = D + H$ is an accounting identity rather than a causal mediation estimator. The retraction changes subsequent directions and lowers raw-dot utility, so it does not identify a free debiasing strategy. Finally, head-item exposure can encode useful priors as well

Table 4: Hybrid serving and utility, means over three seeds. Head values are fractions; lifts and reductions are percentage points. Recall/NDCG show standard→matched-retracted raw-dot runs.

Dataset	Unit/actual head	Tangent-only lift	Standard/retracted norm lift	Paired reduction	Recall@100	NDCG@100
ML-32M	.495/.511	9.87	1.62/−11.98	13.60±.88	.1991→.1644	.2173→.1768
Amazon	.692/.893	18.96	20.12/−.33	20.45±.60	.0398→.0320	.0170→.0133

as unwanted amplification; application-specific relevance, fairness, and business objectives are required to judge it.

6 Conclusion

Cosine-gradient orthogonality is a property of an encoder output; tangent representation motion is a property of the parameter update. A lookup-equivalent item path with direction-preserving row updates bridges the two, making every step add its squared length to item squared norm. When search and recommendation share that path, their update opportunities conditionally accumulate in the same magnitude, which raw-inner-product serving can expose on both surfaces.

Condition tests validate the predicted exact behavior, while a nonlinear tower that combines per-item ID vectors with a shared content encoder shows that a traceable tangent component can remain material outside the guarantee. Its serving effect varies sharply by dataset, and suppressing immediate norm growth reduces utility. The practical result is therefore an audit, not a universal bias or mitigation claim: measure realized motion, separate task-specific opportunity count from energy per update, and decompose direction from magnitude at serving. Joint search–recommendation logs are the necessary next test of cross-surface prevalence.

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